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Grade received 100% To pass 80% or higher

1 / 1 point

1. Suppose we use an unsorted doubly-linked list for the event queue instead of a balanced binary search tree, what is the lowest correct upper bound on the running time of the plane sweep algorithm?

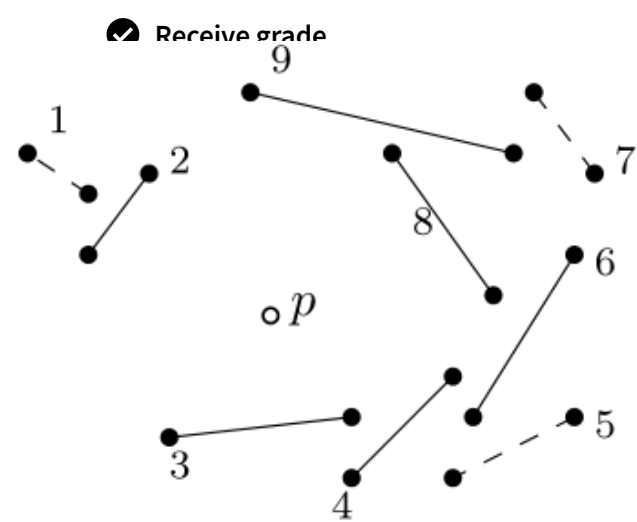
1 / 1 point

- $\wedge$

Correct  
Submit your assignment

Try again

**2.**



Your grade  
100%

**View Feedback** 1 / 1 point

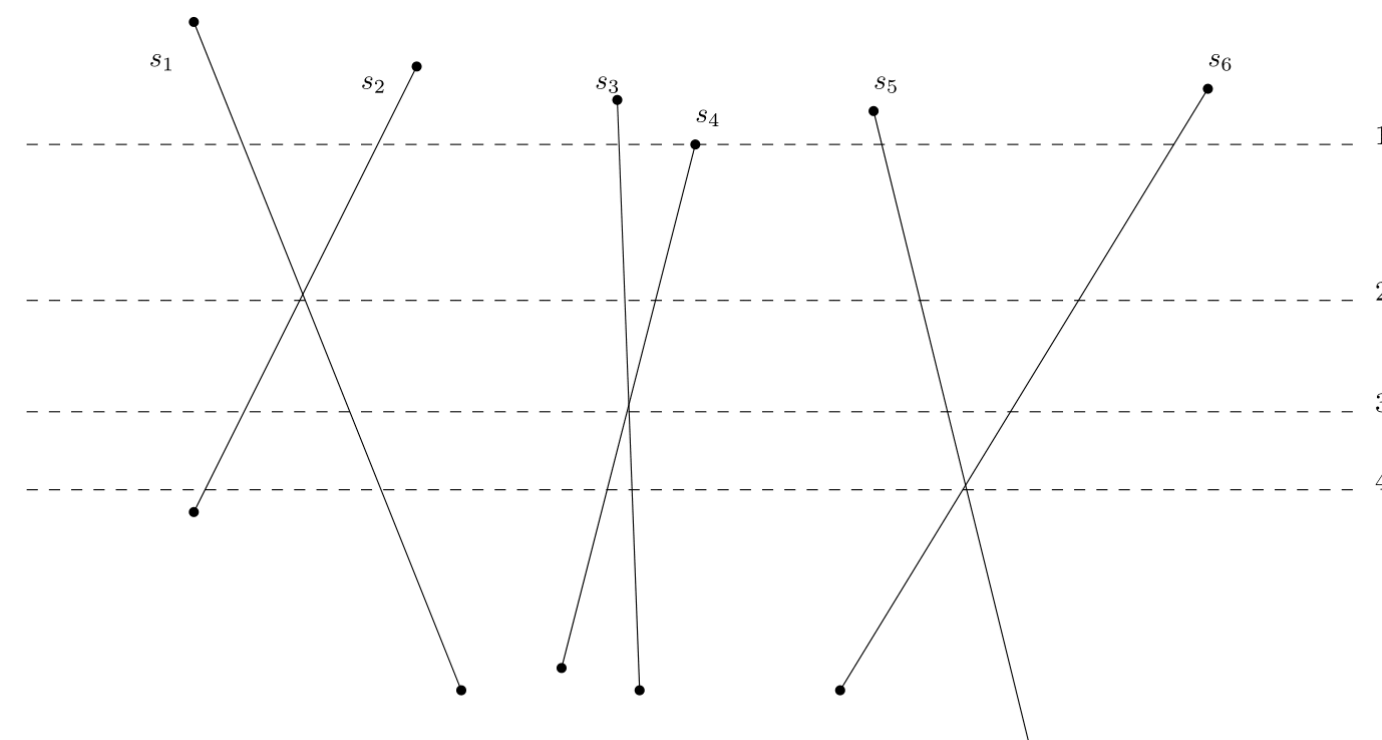
We keep your highest score

Consider a set of disjoint line segments and a point  $p$  not on any of the segments. We wish to determine all segments that  $p$  can see, in the example above the dashed lines are not visible from  $p$ . This problem can be solved using a plane sweep algorithm where the "sweep line" is a ray rotating around  $p$  instead of a vertical sweep line. Which line segments could initially be put in the status structure?

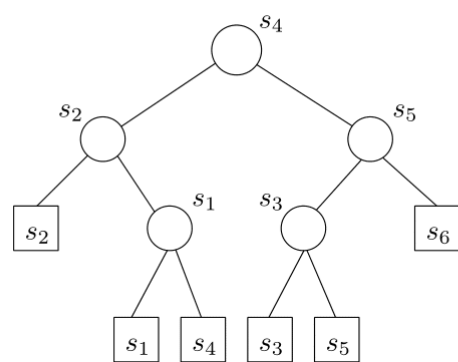
- ☐ Segments 3 and 9.
- ☒ Segments 8 and 9.
- ☐ Segments 3, 4, 5 and 6.

✓ Correct  
Correct, a ray emitted from  $p$  that is rotated hits both segments 8 and 9 at some point.

**3.**



1 / 1 point



Consider the set of line segments in the image above. At the bottom of the image a status structure is displayed. What dashed line does this status structure correspond to? Here we consider the status structures after the event at the dashed line is handled.

- ☐ 1  
☐ 2  
☒ 3  
☐ 4

✔ **Correct**  
Correct

4. What operations does the status data structure need to support?

1 / 1 point

- ☒ Insert - insert a line segment in the status.
- ☒ **Correct**  
Correct, we need to be able to insert line segments in the status.

- ☒ **Delete** - delete a line segment from the status.

✓ **Correct**  
Correct, we need to be able to delete line segments from the status.

- ☒ Search - search for a line segment in the status.

✓ **Correct**  
Correct, we need to be able to search for line segments in the status

- ☐ Traverse - efficiently go through all entries in the status.
- ☐ Update - change properties of a specific element in the status.

- ☑ Successor/predecessor - find the line segment next/previous in the order.

✓ **Correct**  
Correct, for an edge in the status structure we need to check whether it intersects its successor or predecessor in order to report the intersection.

- ☐ Minimum/maximum - find the first/last element in the order.

5. The plane sweep algorithm for line segment intersection first sorts all segment endpoints and inserts them into the event queue. If you ignore the running time of this step, does the remainder of the algorithm run in linear time for an input with  $k$  intersections?

1 / 1 point

- ☒ No, not in general, even if  $k$  is a constant.  
☐ Yes, for any value of  $k$ .  
☐ Yes, if  $k$  is constant.

✓ **Correct**  
Correct, each event still takes logarithmic time to be processed, and there are  $2n + k$  events.